of about 4 or 5 cubic foot per hour. This was completed in the latter end of August 1838, after a few preliminary trials. [16]

The data above is sufficient evidence that a new locomotive engine was constructed. The train, with the locomotive, carriages and passengers, was scheduled for a trial run on 28 August 1838.

INDIA'S FIRST TRAIN FERRYING PEOPLE, 1838

Sir Arthur Thomas Cotton's experiments on rotary steam engines seemed to be making headway. Enough data had been gathered and a new locomotive engine was finally constructed from the variation trials of the stationary steam engine. The new engine was ready for a trial run, with carriages and people, waiting to be ferried on the Red Hills Railway.

Experiment 1: Four-car Train at 2.5 Miles Per Hour

The first experiment was conducted on Tuesday, 28 August 1838. It has been documented under the heading 'Experiment No. 1 with Locomotive'.

The technical specifications of the locomotive were as follows: It had three-foot-long arms [17] with a thickness of 1/5th inch. The weight of the engine and carriage was 2,800 lbs. Three other carriages, with an additional weight of 3,000 lbs, were attached. The coal and water, along with other stuff, added up to about 600 lbs. Twenty-one persons occupied the carriages.

The total weight of the train—including the engine, four carriages, coal and water, and twenty-one persons—totalled up to 8,800 lbs, or 4 tonnes .

The train plied at a speed of about 2.5 miles per hour.

Experiment 2: Two-car Train at 3.5 Miles Per Hour

The second train experiment on the Red Hills Railway was conducted on the same day and it ferried two carriages and ten persons.

It had the same technical specifications as that of the first train, except for the fact that this one possessed a comparatively reduced total weight of 6,500 lbs, since it had just two carriages (weighing about 2,000 lbs) and ten people (weighing about 1,200 lbs).

Being lighter, this train was faster and had a speed of 3.5 miles per hour.